ST443 Week 1

Linear alebgra review

1 October 2026

These slides are not official notes. Always refer to the official material on Moodle

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LINEAR ALGEBRA IN A NUTSHELL

((*The matrix* A *is* n *by* n))

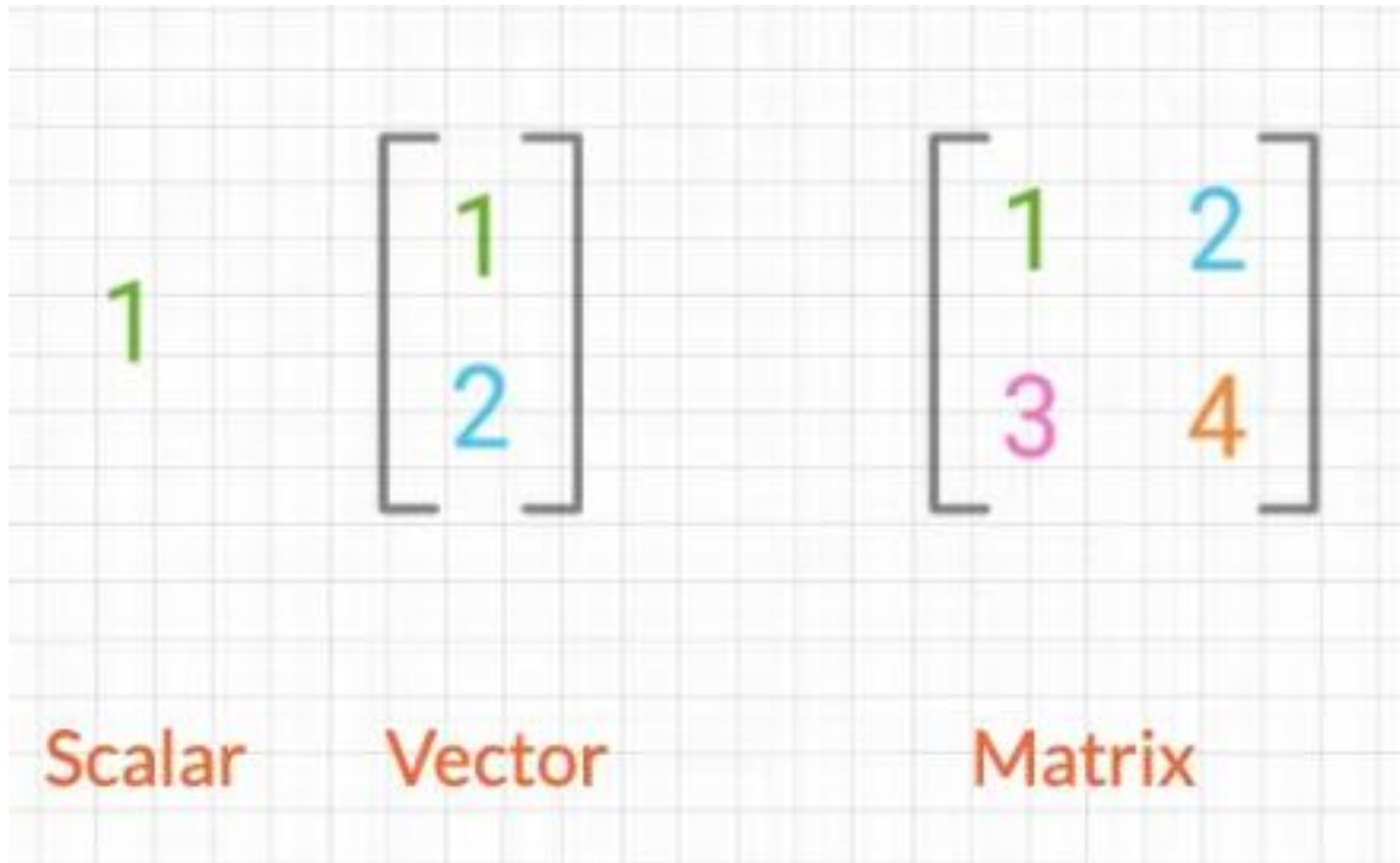
Nonsingular	Singular
A is invertible	A is not invertible
The columns are independent	The columns are dependent
The rows are independent	The rows are dependent
The determinant is not zero	The determinant is zero
$A\boldsymbol{x} = \mathbf{0}$ has one solution $\boldsymbol{x} = \mathbf{0}$	$A\boldsymbol{x} = \mathbf{0}$ has infinitely many solutions
$A\boldsymbol{x} = \boldsymbol{b}$ has one solution $\boldsymbol{x} = A^{-1}\boldsymbol{b}$	$A\boldsymbol{x} = \boldsymbol{b}$ has no solution or infinitely many
A has n (nonzero) pivots	A has $r < n$ pivots
A has full rank $r = n$	A has rank $r < n$
The reduced row echelon form is $R = I$	R has at least one zero row
The column space is all of $\mathbf{R}^n$	The column space has dimension $r < n$
The row space is all of $\mathbf{R}^n$	The row space has dimension $r < n$
All eigenvalues are nonzero	Zero is an eigenvalue of A
$A^{\mathrm{T}}A$ is symmetric positive definite	$A^{\mathrm{T}}A$ is only semidefinite
A has n (positive) singular values	A has $r < n$ singular values

https://jcer.in/jcer-docs/E-Learning/Digital%20Library%20/E-Books/linear-algebra-author-gilbert-strang.pdf

Linear algebra basics

Linear algebra 2

- Matrix transpose of a product
- Matrix inverse: square vs pseudo-inverse
- Matrix multiplication
- Norm: vector and matrix
- Trace: cyclic property, matrix product and trace connection
- Structure matrices


1
1
2
1 2
3 4
Scalar
Vector
Matrix

Some basic types of matrices



$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$$

$$D = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$$



T/F

The transpose of a matrix converts rows to columns, and columns to rows.

Transpose of a square matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Transpose of a rectangular matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

A symmetric matrix is a square matrix that is equal to its transpose.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

$$A = A^T$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + \quad 10$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + 10 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Matrix multiplication

A: m x n matrix

B and C : n x p matrices

D: p x q

T/F

1. $AB = BA$
2. $A(B + C) = AB + AC$
3. $(ABD)^T = D^T B^T A^T$

Definition. An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C satisfying

$$CA = AC = I_n,$$

where I_n is the identity matrix. If it exists, C is called the *inverse* of A, denoted by A^{-1}.

$$\boxed{A^{-1}A = AA^{-1} = I_n}$$

$$\begin{bmatrix} -3 & 1 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{5} \\ 1 & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

matrix A

matrix A^{-1}

2 x 2 identity matrix



T/F

All square matrices are invertible

- A matrix is called nonsingular if it is invertible, and singular if it is non-invertible.



Matrix inversion

If A,B, and C are n x n invertible matrices, then

T/F

1. $(A^{-1})^T = (A^T)^{-1}$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A + B)^{-1} = A^{-1} + B^{-1}$
4. $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

Rank of an m x n matrix A

The word maximum is required

- The *maximum* number of linearly independent row is equal to the *maximum* linearly independent columns. This is the rank of the matrix.

- The rank tells us whether a square matrix has an inverse.



What is the rank of this matrix?

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

Rank of an m x n matrix A

1. rank(A) = rank(A^T)

2. rank($A^T A$) = rank(A)

3. rank(A) = m, we say A is of full row rank; rank(A)=n, A is of full column rank.

4. If rank(A) = $m = n$, we say the matrix has full rank, and A has a unique inverse.

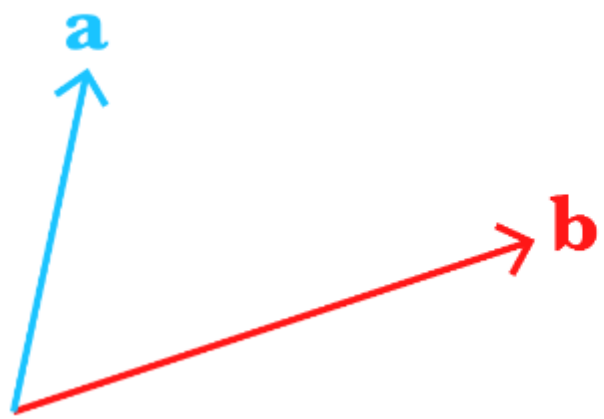


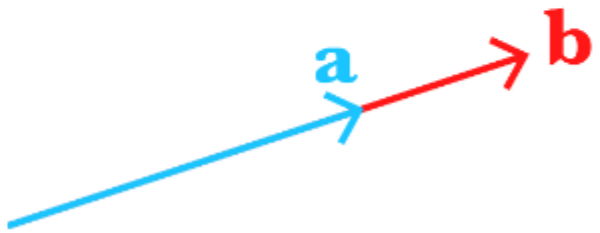
T/F

rank (A) $\leq \min(m, n)$

A collection of vectors are linearly independent if the only way to make them equal to the zero-vector is if all the coefficients are zero:

$$\mathbf{0} = \alpha_1 \mathbf{a}_1 + \cdots + \alpha_n \mathbf{a}_n \quad \Longleftrightarrow \quad \alpha_1 = \cdots = \alpha_n = 0.$$





Column space

If $\mathbf{A} = [\mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n]$ where $\mathbf{a}_i$ are the columns, then a vector $\mathbf{b}$ lies in $\mathrm{Col}(\mathbf{A})$ if:

$$\mathbf{b} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n$$

for some scalars $c_1, c_2, \ldots, c_n$.

Example:

Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

The columns are $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$

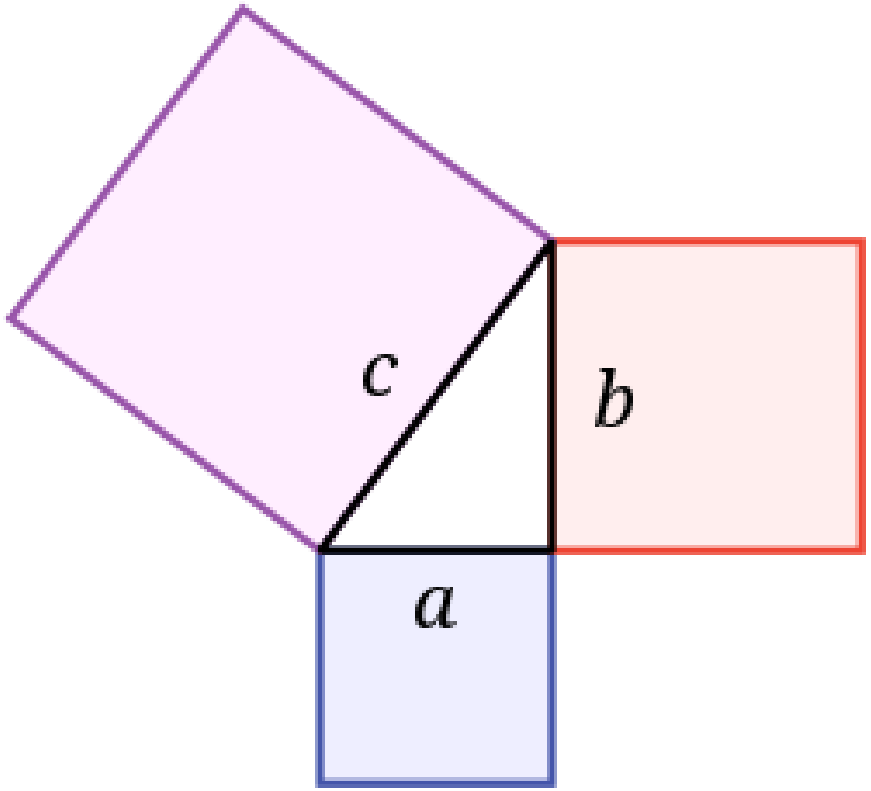
Does $\mathbf{b} = \begin{bmatrix} 5 \\ 11 \\ 17 \end{bmatrix}$ **lie in** $\mathrm{Col}(\mathbf{A})$?

$$\mathbf{b} = 1 \cdot \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + 2 \cdot \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \\ 17 \end{bmatrix}$$

Distance metrics: Pythagoras' theorem

Pythagoras' theorem: the area of the square whose side is the hypotenuse (the side opposite the right angle) is equal to the sum of the areas of the squares on the other two sides

$$a^2 + b^2 = c^2$$



Norm

- A norm is a function that measures the size or magnitude of a vector or matrix.

- The Euclidean norm is a commonly used norm in ML. It is the generalisation of Pythagoras' theorem to find the length (norm) of an n-dimensional vector

Euclidean norm

- $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ $n \times 1$ vector

- $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}$ a scalar

- $\|x\|_2^2 = x^T x$

$\|x\|_2$: the 2 ↑

- Without this subscript, it conventionally denotes the Euclidean norm.
- If working with a number of different norms, we should include the subscript.



Name the two fundamental problems in linear algebra

Two fundamental problems in linear algebra

A. Solving systems of linear equations

Given an m x n matrix A and a vector b, find the vector x that solves it:

$$\text{Solve } Ax = b$$

B. Understanding the behaviour of linear transformations

Given a square matrix A, find scalars λ (eigenvalues) and vectors x (eigenvectors) such that:

$$Ax = \lambda x \text{ , where } x \neq 0$$

https://math.emory.edu/~lchen41/teaching/2020_Fall/Nicholson-OpenLAWA-2019A.pdf

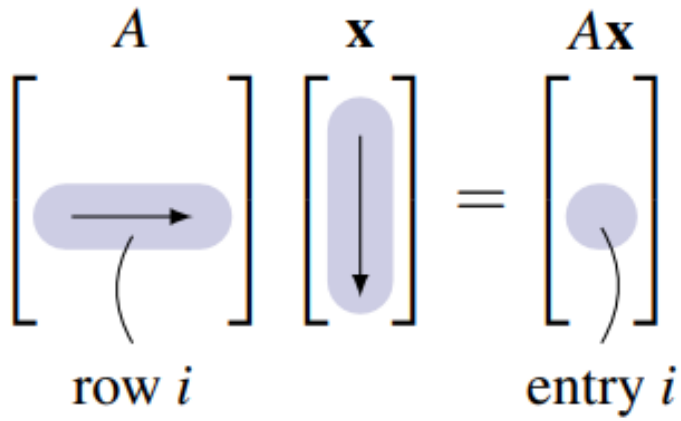
Solving systems of linear equations

Example. Solve the system of 2 linear equations with 4 unknowns.

$$\begin{aligned} x_1 - 2x_2 + 3x_3 + x_4 &= -3 \\ 2x_1 - x_2 + 3x_3 - x_4 &= 0 \end{aligned}$$

We can write the system of equations in the fowm $Ax = b$

$$\begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$$



$$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} + x_3 \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} + x_4 \begin{bmatrix} a_{14} \\ a_{24} \\ a_{34} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 \end{bmatrix}$$

Example.

$$\mathbf{Ax} = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2\cdot 2 + (-1)1 + 3\cdot 0 + 5(-2) \\ 0\cdot 2 + 2\cdot 1 + (-3)0 + 1(-2) \\ (-3)2 + 4\cdot 1 + 1\cdot 0 + 2(-2) \end{bmatrix} = \begin{bmatrix} -7 \\ 0 \\ -6 \end{bmatrix}$$

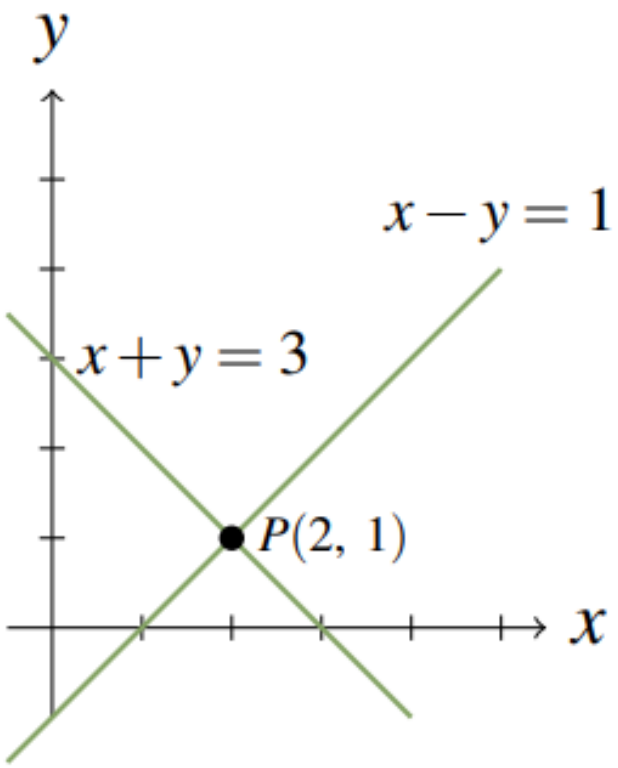


T/F

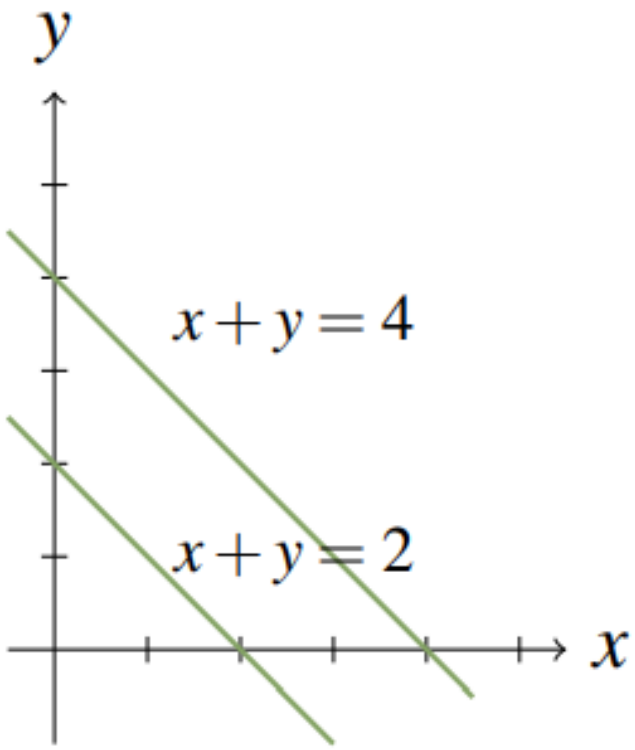
A system of linear equations has one of:

- Unique solution
- No solution
- Infinitely many solutions

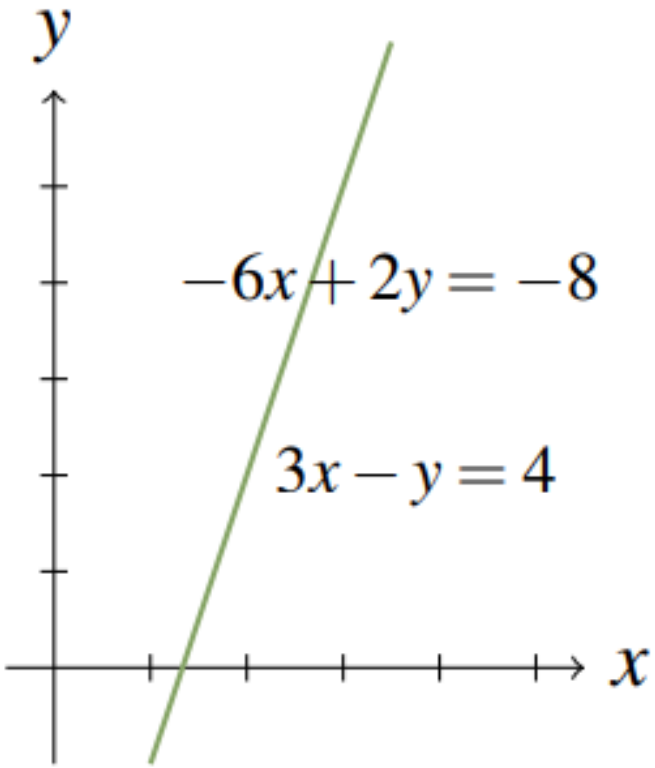
Example. 2 linear equations with 2 unknowns



(a) Unique Solution
$(x = 2, y = 1)$



(b) No Solution



(c) Infinitely many solutions
$(x = t, y = 3t - 4)$

Methods to solve Ax = b

- Gaussian Elimination / Row reduction
- LU Decomposition
- Cramer's rule
- Cholesky decomposition (for symmetric positive definite matrices)

Software packages like NumPy and MATLAB choose the appropriate method based on matrix properties

Understanding the behaviour of linear transformations

$$A\vec{x} = \lambda\vec{x}$$

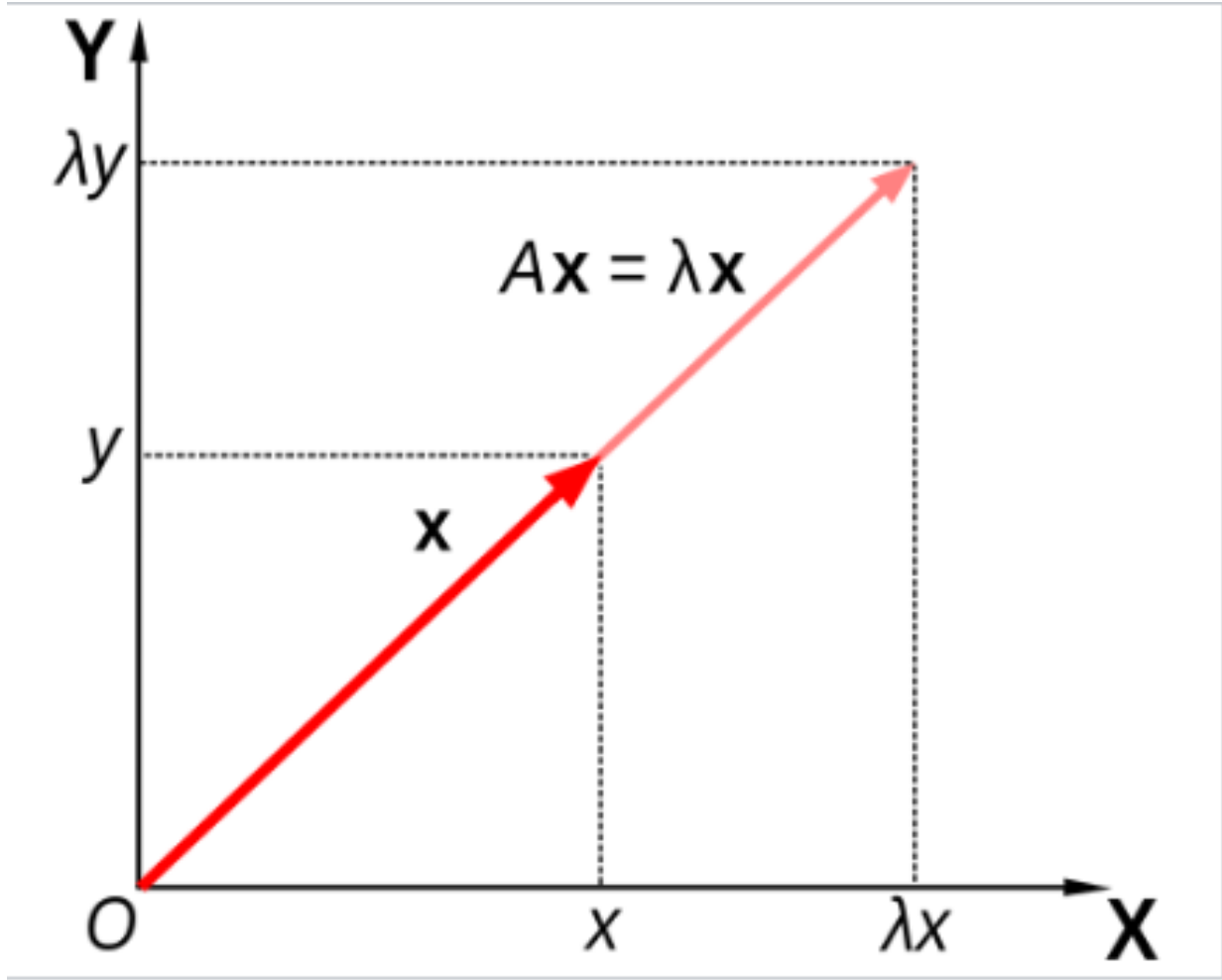
$\vec{x}$ = eigenvector

λ = eigenvalue

each eigenvalue is associated with a specific eigenvector

Example 3.3.2

If $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ then $A\mathbf{x} = 4\mathbf{x}$ so $\lambda = 4$ is an eigenvalue of A with corresponding eigenvector $\mathbf{x}$.



Matrix A acts by stretching the vector $\mathbf{x}$, not changing its direction, so $\mathbf{x}$ is an eigenvector of A.



How can we find the eigenvalues of a matrix?

Characterstic polynomial

Definition. Let A be an $n \times n$ matrix. The ***characteristic polynomial*** of A is the function $f(\lambda)$ given by

$$f(\lambda) = \det(A - \lambda I_n).$$

Theorem (Eigenvalues are roots of the characteristic polynomial). *Let A be an $n \times n$ matrix, and let $f(\lambda) = \det(A - \lambda I_n)$ be its characteristic polynomial. Then a number λ_0 is an eigenvalue of A if and only if $f(\lambda_0) = 0$.*

https://textbooks.math.gatech.edu/ila/characteristic-polynomial.html

Characterstic **polynomial**

Theorem. *Let A be an $n \times n$ matrix, and let $f(\lambda) = \det(A - \lambda I_n)$ be its characteristic polynomial. Then $f(\lambda)$ is a polynomial of degree n. Moreover, $f(\lambda)$ has the form*

$$f(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} \operatorname{Tr}(A) \lambda^{n-1} + \cdots + \det(A).$$

In other words, the coefficient of λ^{n-1} is $\pm \operatorname{Tr}(A)$, and the constant term is $\det(A)$ (the other coefficients are just numbers without names).

Find the characteristic polynomial of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Solution

We have

$$f(\lambda) = \lambda^2 - \operatorname{Tr}(A)\lambda + \det(A) = \lambda^2 - (5+1)\lambda + (5 \cdot 1 - 2 \cdot 2) = \lambda^2 - 6\lambda + 1,$$

Positive definite matrix

A square real matrix A is positive definite if $v^T Av > 0$ for all $v \neq 0$



How to determine if this matrix positive definite?

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$



T/F

A matrix is positive definite if and only if it is symmetric and all of its eigenvalues are strictly positive



T/F

Let A be an n x n matrix. The following statements are equivalent :

1. A is invertible
2. 0 is not an eigenvalue of A
3. The columns of A are linearly independent.

Invertible Matrix Theorem. *Let A be an $n \times n$ matrix, and let $T : \mathbf{R}^n \to \mathbf{R}^n$ be the matrix transformation $T(x) = Ax$. The following statements are equivalent:*

1. *A is invertible.*
2. *A has n pivots.*
3. $\mathrm{Nul}(A) = \{0\}$.
4. *The columns of A are linearly independent.*
5. *The columns of A span $\mathbf{R}^n$.*
6. *$Ax = b$ has a unique solution for each b in $\mathbf{R}^n$.*
7. *T is invertible.*
8. *T is one-to-one.*
9. *T is onto.*
10. $\det(A) \neq 0$.
11. *0 is not an eigenvalue of A.*

https://textbooks.math.gatech.edu/ila/eigenvectors.html

Exercise in linear algebra: derive the OLS estimator

Consider the linear regression model:

$$y = X\beta + \varepsilon$$

Show that the OLS estimator is given by

$$\hat{\beta}^{\mathrm{OLS}} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$$

whenever X has full column rank

$$\beta = (\beta_1, \ldots, \beta_p)^\top \in \mathbb{R}^p$$

p x 1 (column) vector
of unknown parameters

$y \in \mathbb{R}^n$

n x 1 vector

$$y = X\beta + \varepsilon$$

$\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^\top$

n x 1 (column) vector of error terms
i.i.d. variables with mean 0, variance σ^2

$$X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n \times p}$$

X is a n x p matrix

n =... p = ...

Each column of X is a column vector

- Bivariate/simple linear regression with intercept has p=2
- Multiple linear regression has p>2).

$$y = X\beta + \varepsilon$$

	y	x
	10	2
	15	7
	12	4



$$10 = \beta_0 + 2\beta_1 + \epsilon_1,$$
$$15 = \beta_0 + 7\beta_1 + \epsilon_2,$$
$$12 = \beta_0 + 4\beta_1 + \epsilon_3.$$

$$\mathbf{y} = \begin{pmatrix} 10 \\ 15 \\ 12 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 2 \\ 1 & 7 \\ 1 & 4 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

n = ...

p =...

Linear algebra basics

Linear algebra 2



What is the trace of matrix A?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$



A: $m \times n$ matrix

B: n $\times m$ matrix

T/F

$$tr(AB) = tr(BA)$$

A: $m \times n$ matrix

B: n $\times$ m matrix

$$tr(AB) = \sum_{i=1}^{m} (AB)_{ii}$$

A: $m \times n$ matrix

B: n $\times$ m matrix

$tr(AB) = \sum_{i=1}^{m} (AB)_{ii}$

Fill in the XX

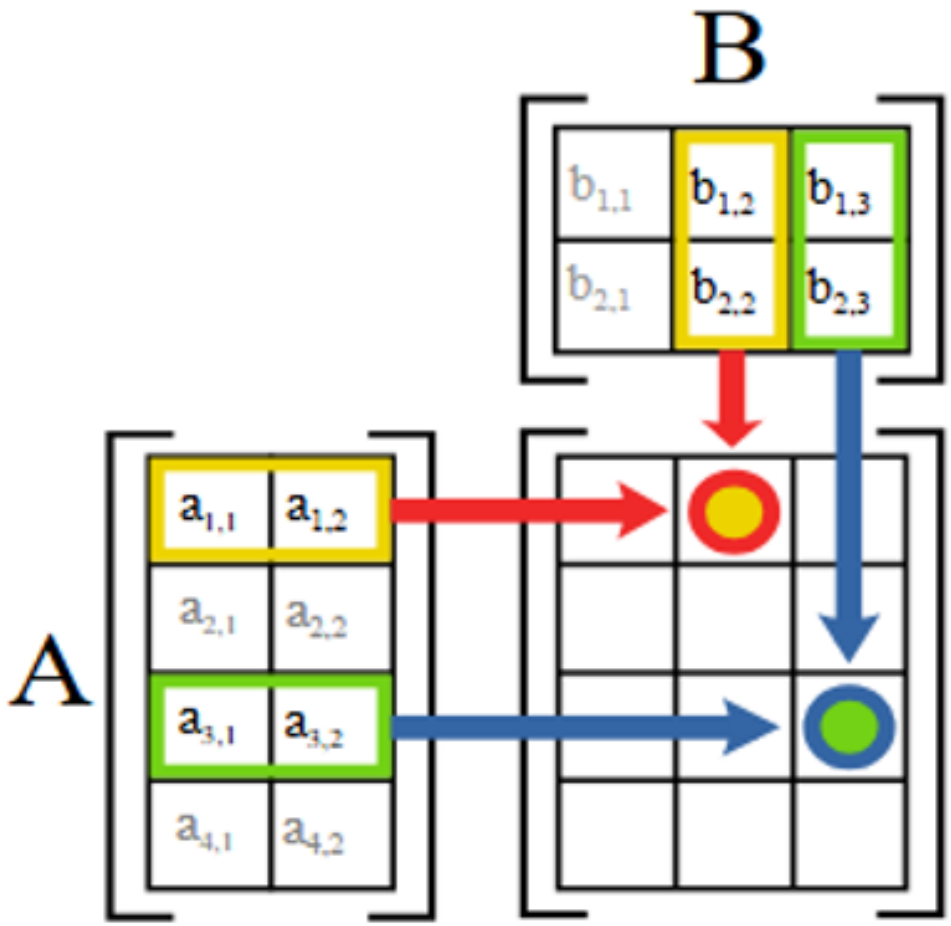
$$(AB)_{ii} = \sum_{j=1}^{n} A_{XX} B_{XX}$$

A: $m \times n$ matrix

B: n $\times$ m matrix

$$tr(AB) = \sum_{i=1}^{m} (AB)_{ii}$$

$$(AB)_{ii} = \sum_{j=1}^{n} A_{ij} B_{ji}$$



A: $m \times n$ matrix

B: n $\times$ m matrix

$$tr(AB) = \sum_{i=1}^{m} (AB)_{ii}$$

$$= \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ji}$$

$$= \sum_{j=1}^{n} \sum_{i=1}^{m} B_{ji} A_{ij}$$



A: $m \times n$ matrix; B: n $\times$ m matrix

$$tr(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ji}$$

$$= \sum_{j=1}^{n} \sum_{i=1}^{m} B_{ji} A_{ij}$$

Swapping the double sum

$$= tr(BA)$$



Cyclic property of trace

T/F

For matrices with compatible dimensions, in general:

$$tr(ABC) = tr(ACB)$$

Frobenius norm

The Frobenius norm is a matrix norm - it measures the magnitude of a matrix.

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})}$$

Example

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Sum of squares $= 1 + 4 + 9 + 16 = 30.$

So $\|A\|_F = \sqrt{30}.$



$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})}$$

Does this measure have a similarity with the Euclidean norm for vectors?

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})}$$

Interpretation

- The Frobenius norm of a matrix is the equivalent of the Euclidean norm for vectors.

- It’s the Euclidean distance when you flatten the matrix into a vector.



Frobenius norm is orthogonally invariant

What does this statement mean?

Frobenius norm is orthogonally invariant

The Frobenius norm remains unchanged under multiplication by orthogonal matrices from the left, right, and hence on both sides.

Frobenius norm is orthogonally invariant

Example

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Sum of squares $= 1 + 4 + 9 + 16 = 30.$

So $\|A\|_F = \sqrt{30}.$

orthogonal rotation matrix (90°)

$$Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad Q^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

$Q^T Q = I.$

$$QA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} -3 & -4 \\ 1 & 2 \end{pmatrix}.$$

$$\|QA\|_F = \sqrt{30} = \|A\|_F$$

$$AQ = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 4 & -3 \end{pmatrix}.$$

$$\|AQ\|_F = \sqrt{30} = \|A\|_F$$

$$QAQ^T = \begin{pmatrix} -3 & -4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}$$

$$\|QAQ^T\|_F = \sqrt{30} = \|A\|_F$$

Structure matrices

- Structure matrices: matrices with properties that can be exploited for computational efficiency (multiplication, inverse,..)

- Examples of the structures
  - Symmetric, diagonal, orthogonal structures
  - Sparsity-based: most entries are zero
  - Rank structure (rank is less than # rows/cols)

  ...and many more.

Orthogonal matrix

- A square matrix is Q is orthogonal if

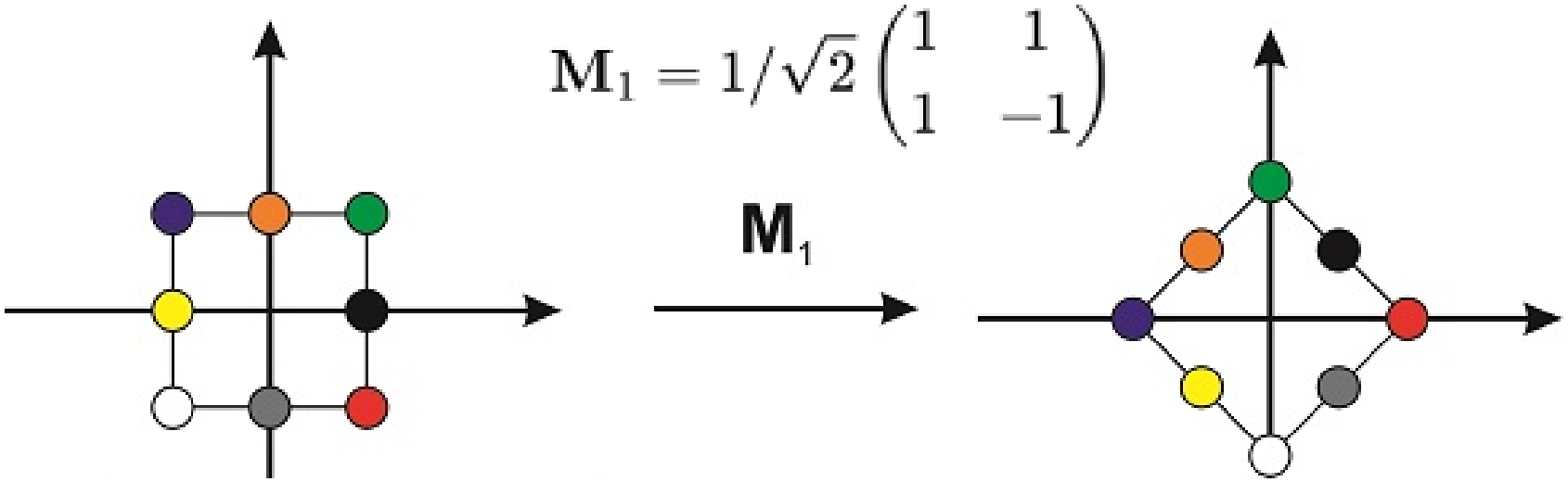
  $Q^T Q = Q Q^T = I$ i.e. Q has orthonormal columns

  Equivalently: $Q^{-1} = Q^T$

Orthogonal matrix

- Orthogonal matrices perform rigid transformations:-
  - All geometric properties are preserved:
    distances, angles, lengths, shape, and volumes

  - Represent rotations or rotation + reflection



Orthogonal diagonalisation

A square matrix A is said to the orthogonally diagonalisable when an orthogonal matrix Q and a diagonal matrix D can be found such that

$$A = QDQ^T$$

https://math.emory.edu/~lchen41/teaching/2020_Fall/Section_8-2.pdf

Orthogonal diagonalisation

- An n x n matrix is orthogonally diagonalisable if and only if A is symmetric



T/F

The singular value decomposition (SVD) extends the orthogonal diagonalisation for symmetric matrices to work for any matrix.

Singular value decomposition (SVD)

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate) , where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$. (intuition? general form? Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.
- $\mathbf{\Sigma} = \text{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries, $\searrow$.
- σ_i: singular values.

Computing the SVD

$$A = U\Sigma V^T$$

- The ordering of the vectors comes from the ordering of the singular values (largest to smallest).
- The columns of U are the eigenvectors of AA^T
- The columns of V are the eigenvectors of A^TA.
- The diagonal elements of Σ are the singular values, $\sigma_i = \sqrt{\lambda_i}$

http://people.whitman.edu/~hundledr/courses/M240S16/M240/TheSVD.pdf

Low rank matrices

- An $m \times n$ matrix A is low rank if the rank of A is much smaller than both dimensions.
  - Its rows/columns are linearly dependent
  - Lots of redundancy – most vectors are linear combinations of the few basis vectors.

Low-rank approximation

- The matrix we have may be too large to store or work with.

- Aim: The best rank-k approximation to matrix A - a complex, high-rank matrix – is a simpler, k rank (low-rank) matrix that captures the essential structure of A.
  - The lower the k, the more compression, but error increases.

https://web.stanford.edu/class/cs168/l/l9.pdf

Low-rank approximation

Objective: constrained optimisation problem

$$\min_{\hat{\mathbf{S}} \in \mathbb{R}^{4\times 4}: \operatorname{rank}(\hat{\mathbf{S}})=2} \left\| \mathbf{S} - \hat{\mathbf{S}} \right\|_{\mathrm{F}}$$

Among all 4x4 matrices of rank 2, Find the one that is closest in the sense of the Frobenius norm to S

Solution: (Eckart-young-Mirsky theorem) Solution is given by a truncated SVD keeping the top k singular values and vectors.



T/F

A. Every matrix has an inverse
B. Every matrix has a unique pseudo-inverse

Pseudo-inverse: summary

- **Pseudoinverse** $\mathbf{A}^\dagger \in \mathbb{R}^{n\times m}$ for a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$:
  - *Definition*: $\mathbf{A}\mathbf{A}^\dagger\mathbf{A} = \mathbf{A}$, and $\mathbf{A}^\dagger\mathbf{A}\mathbf{A}^\dagger = \mathbf{A}^\dagger$, and both $\mathbf{A}\mathbf{A}^\dagger, \mathbf{A}^\dagger\mathbf{A}$ are symmetric
  - *Existence*: $\mathbf{A}^\dagger$ always exists and is unique
  - *Computing*: How to find the pseudoinverse:
    - If $\mathbf{A}$ has full column rank: $\mathbf{A}^\dagger = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$
    - If $\mathbf{A}$ is "diagonal": $\mathbf{A}^\dagger \in \mathbb{R}^{n\times m}$ is also "diagonal" with reciprocals of nonzero diagonals of $\mathbf{A}$
    - In general: $\mathbf{A}^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T$ (if $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$)

https://www.sjsu.edu/faculty/guangliang.chen/Math250/lec6ginverse.pdf

Pseudo-inverse If $\mathbf{A}$ has full column rank: $\mathbf{A}^\dagger = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$

Find the pseudoinverse of $\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution: Observe that this matrix has full column rank (i.e., 2).

Since

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

we have

$$\mathbf{A}^\dagger = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \end{pmatrix}$$

Pseudo-inverse $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a diagonal matrix

Theorem 0.6. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a diagonal matrix, i.e., all of its entries are zero except some of those along its diagonal. Then the pseudoinverse of $\mathbf{A}$ is another diagonal matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ such that

$$b_{ii} = \begin{cases} \frac{1}{a_{ii}}, & \text{if } a_{ii} \neq 0 \\ 0, & \text{if } a_{ii} = 0 \end{cases}$$

Proof. We verify this result using an example. Let

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Pseudo-inverse

$\mathbf{A} \in \mathbb{R}^{m \times n}$ be an arbitrary matrix.

Theorem 0.7. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be an arbitrary matrix. Suppose its full SVD is $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. Then the pseudoinverse of $\mathbf{A}$ is

$$\mathbf{A}^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T$$

Example. Computing pseudoinverse by SVD

$$\underbrace{\begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\mathbf{A}} = \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{\mathbf{U}_2} \cdot \underbrace{\begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{\Sigma}_2} \cdot \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}^T}_{\mathbf{V}_2^T}$$

By the last theorem,

$$\mathbf{A}^\dagger = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{\mathbf{V}_2} \cdot \underbrace{\begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{\Sigma}_2^{-1}} \cdot \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}^T}_{\mathbf{U}_2^T} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \end{pmatrix}$$

Pseudo-inverse: usefulness – an example

Problem: solution of

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

might not exist or may not be unique.

The pseudo-inverse based solution $\hat{\mathbf{x}} = \mathbf{A}^{-}\mathbf{b}$ handles this issue and it is 'optimal'.

- smallest error: for any $\mathbf{x} \in \mathbb{R}^n$

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2 \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|_2.$$

- smallest norm: among the $\mathbf{x}$ vectors for which '$=$' holds $\hat{\mathbf{x}}$ has minimal Euclidean norm.

Derivation of OLS for linear regression

$$\frac{\partial}{\partial \beta}\|\mathbf{Y}-\mathbf{X}\beta\|^2 = \frac{\partial}{\partial \beta}(\mathbf{Y}-\mathbf{X}\beta)^\top(\mathbf{Y}-\mathbf{X}\beta)$$

$$= \frac{\partial}{\partial \beta}(\mathbf{Y}^\top\mathbf{Y} - 2\beta^\top\mathbf{X}^\top\mathbf{Y} + \beta^\top\mathbf{X}^\top\mathbf{X}\beta)$$

$$\frac{\partial(\boldsymbol{a}'\boldsymbol{x})}{\partial \boldsymbol{x}} = \frac{\partial(\boldsymbol{x}'\boldsymbol{a})}{\partial \boldsymbol{x}} = \boldsymbol{a}$$

$$\frac{\partial(\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x})}{\partial \boldsymbol{x}} = 2\boldsymbol{A}\boldsymbol{x}$$

for symmetric A

Solution to FOC is given by the OLS estimator $\hat{\beta}$

$$0 = -2\mathbf{X}^\top\mathbf{Y} + 2\mathbf{X}^\top\mathbf{X}\hat{\beta}$$

$$0 = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\beta}$$

Rearrange to give a system of linear equations called the normal equations:

$$\underset{A}{\mathbf{X}^\top \mathbf{X}} \underset{x}{\hat{\beta}} = \underset{b}{\mathbf{X}^\top \mathbf{Y}}$$

$$\underbrace{\begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}}_{b}$$